SAI RAM SAGAR

Software Developer · Real-Time & Geospatial Web Applications · React.js / Node.js

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SUMMARY

Software Developer with 2+ years on a statewide emergency-response platform (ERSS) serving 700+ police stations across Telangana and handling 6,000-10,000 citizen calls a day. Primarily frontend - real-time React.js operator interfaces and Leaflet geospatial workflows - with backend work on a sub-second Speech-to-Text pipeline in Node.js, WebSocket and Pusher.

TECHNICAL SKILLS

Frontend: JavaScript (ES6+), React.js, HTML5, CSS3, Tailwind CSS, Axios

Backend & Data: Node.js, Express.js, REST APIs, WebSocket, MongoDB, MySQL, SQLite

Real-Time: Event-Driven Architecture, Pusher, WebSocket, State Synchronisation, REST API Integration

Geospatial: Leaflet, Geofencing & Polygon Rendering, Map Tiles, Self-Hosted Tile Sources

Speech & Audio: Speech-to-Text (STT), Voice Activity Detection (VAD), 8 kHz-16 kHz Resampling, Buffering, Speech Segmentation

Tooling & Practices: Git, GitHub, PM2, Bugsnag, SonarQube, npm, Render, Netlify, AI-assisted & CLI-based development, Technical Documentation

PROFESSIONAL EXPERIENCE

Software Developer — Krishnas Digital Technologies and Softlabs Pvt. Ltd. **May 2024 - Present**

Hyderabad, India · Emergency Response Support System (ERSS) for Telangana Police

- **Built real-time Computer-Aided Dispatch (CAD) workflows** in React.js for SOS alerts, emergency calls, service assignment and live incident data, used by up to 50 dispatch operators.
- **Developed a real-time Speech-to-Text pipeline** (telephony audio to VAD to WebSocket to STT to Pusher to CAD UI) delivering live call transcripts to operators in under 1 second, with 8 kHz to 16 kHz resampling and speech segmentation.
- **Built live vehicle tracking and police-station geofencing** on Leaflet across 700+ stations statewide, keeping 100-200 concurrently active complaints visible on the map using zoom-based rendering, multiple tile sources and tile caching.
- **Implemented Pusher/WebSocket event flows** that removed page refreshes for alerts, vehicle positions and live transcripts, keeping operator screens current during active incidents.
- **Built and maintained the shared React component layer** - data tables, complaint-view modals, popups, animations and state patterns - reused across 50+ modules in the platform.
- **Consolidated duplicated logic into shared utilities** - for example replacing hard-coded map-initialisation code across modules with one parameterised function - guided by SonarQube findings on code smells and technical debt.
- **Diagnosed and fixed production issues** including map-related render crashes, dropped WebSocket connections, API timeouts and async error handling, using Bugsnag and PM2.
- **Used AI-assisted and CLI-based development workflows** to accelerate delivery and evaluate implementation alternatives while retaining ownership of code quality.

KEY PROJECTS

Drone as First Responder (DFR)

React.js · Node.js · Express.js · REST APIs · Leaflet · Geospatial Data

- Building a mission-planning system that lets operators create, configure and reuse predefined drone flight missions.
- Implemented a Flight Hub interface for ordered routing through unlimited waypoints (bounded in practice by drone battery endurance), with add, edit, delete and reposition actions and live path rendering.
- Contributing to design discussions on mission planning, drone assignment, dispatch integration and geospatial visualisation.

Real-Time Speech-to-Text System

Node.js · WebSocket · STT · VAD · Pusher · Audio Processing

- Built the pipeline from telephony audio capture through to live transcript delivery inside the CAD application, at sub-second latency.
- Implemented independent VAD processing for agent and citizen audio streams with configurable speech and silence thresholds.
- Handled 8 kHz to 16 kHz conversion, buffering and speech segmentation, with error handling and observability at each stage.

EDUCATION & CERTIFICATION

B.E. - Vasavi College of Engineering, Hyderabad (2023)

NxtWave - Software Developer Course Completion Certificate